

## Stack & Queue Implementation using Structures (Self-Learning)

Q. Why is it advantageous to implement Stacks and Queues using Structures in C? Provide a C program demonstrating the implementation of a Stack and a Queue where their components are encapsulated within a structure.

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**\*\*Definition to Remember\*\***

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### 1. Advantages of Using Structures

\* **Encapsulation:** Data array and control variables are logically grouped together.

### 2. Stack Implementation using Structure

```
#include <stdio.h>
```

```
// Define the Stack Structure
```

```
// Initialize stack
```

```
// Push operation
```

### 3. Queue Implementation using Structure

```
#include <stdio.h>
```

```
// Define the Queue Structure
```

```
// Initialize queue
```

```
void init
```

```
// Enqueue operation
```

```
void en
```

```
---
```

***\*\*Must-Write Points (for 10 marks)\*\****

***1. Stand***

```
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```

***\*\*Quick Recall\*\****

***`Implem  
(s1, s2)***